

NeuroDual-Net: A Multi-Task Learning Framework for Explainable Brain Tumor Classification and Segmentation[★]

ARTICLE INFO

Keywords:

Brain Tumor Classification
Multi-Task Learning
Medical Image Segmentation
Deep Learning
MRI Analysis
NeuroDual-Net

ABSTRACT

Brain tumor diagnosis from MRI scans typically requires both accurate classification of tumor types and precise segmentation of tumor boundaries. Traditional approaches either treat these as separate tasks or rely on single-task architectures that lack interpretability. We propose NeuroDual-Net, a novel multi-task learning framework that simultaneously performs brain tumor classification and segmentation. The architecture employs a shared encoder backbone with dual decoder heads: a classification head for tumor type identification (glioma, meningioma, pituitary, no tumor) and a segmentation head for tumor boundary delineation. We incorporate Multi-Level Supervision (MLS) at intermediate decoder stages to enhance gradient flow and feature learning. Additionally, we systematically evaluate data augmentation strategies and benchmark against 10 state-of-the-art architectures including EfficientNet-B0, ResNet50, ConvNeXt-Tiny, ViT-Base, Swin-Tiny, MobileNetV3-Large, and SResNet50. NeuroDual-Net achieves 98.60% classification accuracy on the BRISC 2025 dataset, outperforming all benchmarked architectures including Vision Transformers and ConvNeXt variants. The model attains a Dice score of 0.905 for segmentation, with statistical significance confirmed through paired t-tests ($p < 0.05$). Ablation studies demonstrate that the multi-task formulation improves classification accuracy by 6.7% over single-task classification, while data augmentation provides consistent performance gains across all metrics. The multi-task architecture not only achieves superior accuracy but also provides inherent interpretability through joint segmentation outputs.

1. Introduction

Brain tumors are among the most critical neurological disorders, requiring early and accurate diagnosis to improve patient survival and treatment planning. Magnetic Resonance Imaging (MRI) is widely used for brain tumor analysis due to its superior soft tissue visualization and non-invasive imaging capability. However, manual interpretation of MRI scans is time-consuming, subjective, and highly dependent on radiological expertise, particularly when dealing with heterogeneous tumor structures and ambiguous tumor boundaries [1, 2]. Consequently, automated deep learning-based approaches have emerged as promising solutions for reliable brain tumor classification and segmentation [3, 4].

Recent advances in convolutional neural networks (CNNs) and transformer-based architectures have significantly improved MRI-based tumor analysis [5]. Thahiruddin and Wulandari [6] conducted a comparative study of CNN and hybrid transformer models on the BRISC dataset, demonstrating that MobileViTV2-100, ResNet18, and EfficientNet-B0 achieved state-of-the-art classification accuracy exceeding 98%. Their findings highlighted the effectiveness of transformer-enhanced architectures for capturing both local and global contextual information in MRI scans. Similarly, Esfahani et al. [7] proposed NeXtSwin-X, a dual-branch fusion architecture combining ConvNeXt and Swin Transformer backbones with cross-attention mechanisms, achieving highly accurate brain tumor classification while reducing training overhead.

The availability of large-scale annotated datasets has also accelerated research progress in this domain and the method like Ensemble Feature Selection (EFS) have become an essential step [8]. Fateh et al. [9] introduced the BRISC dataset,

a publicly available benchmark comprising 6,000 contrast-enhanced T1-weighted MRI scans with expert-annotated segmentation masks. The dataset includes glioma, meningioma, pituitary tumor, and no-tumor classes across axial, sagittal, and coronal views, enabling comprehensive evaluation for both classification and segmentation tasks. The introduction of BRISC has encouraged the development of robust multi-task and cross-view learning frameworks for clinical brain tumor analysis.

Traditional CNN-based methods have demonstrated strong performance in brain tumor classification tasks [10, 11]. Mzoughi et al. [12] proposed a deep multi-scale 3D CNN for glioma classification using volumetric MRI scans, showing the effectiveness of 3D convolutional feature extraction for tumor representation learning. Rehman et al. [13, 14] further improved classification performance through a hybrid 3D CNN framework integrated with feature selection techniques for microscopic brain tumor detection. TimeDistributed CNN-LSTM architectures have also been explored for exploiting temporal and spatial dependencies in 3D MRI volumes, achieving promising performance for multi-class tumor classification [12].

Beyond classification, tumor segmentation plays a crucial role in treatment planning, surgical navigation, and radiotherapy [15]. Karayegen and Aksahin [16] proposed a semantic segmentation framework using deep learning and 3D imaging techniques for accurate tumor localization. Sun et al. [17] introduced a multi-pathway 3D fully convolutional network (FCN) for multimodal brain tumor segmentation, enabling effective extraction of tumor sub-regions from MRI images. Nguyen et al. [18] further enhanced segmentation performance by integrating an auxiliary classification network into the segmentation framework, demonstrating

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that classification-guided segmentation can improve feature learning and tumor boundary delineation.

Recent studies have increasingly focused on hybrid architectures integrating CNNs, transformers, and attention mechanisms [19]. Abbas et al. [20] proposed a transformer-based encoder-decoder fusion framework for brain tumor segmentation, combining a SegFormer encoder with an attention-enhanced decoder to achieve highly accurate boundary delineation. Bindhu and Vimala [21] developed A-Lite UACNet, a lightweight hybrid attention U-Net architecture inspired by active contour models for efficient tumor segmentation and classification under limited computational budgets. Similarly, Singh et al. [22] introduced EffiNet-4HA, which combines EfficientNetV2-S with Multi-Head Self-Attention (MHSA) modules to enhance contextual feature representation and classification accuracy, while [23] presents EFFResNet-ViT, a novel hybrid deep learning (DL) model.

Transfer learning and ensemble learning approaches have also shown remarkable effectiveness for MRI-based brain tumor analysis [24]. Saha et al. [25] investigated custom ensemble frameworks using DenseNet201, InceptionV3, and Xception models combined with meta-learning strategies, achieving near-perfect classification performance on the BRISC dataset. Mahmud et al. [26] conducted a comparative study of multiple pre-trained CNN architectures with k-fold cross-validation and ensemble learning, reporting superior classification accuracy and robust generalization across MRI views. Furthermore, preprocessing-enhanced CNN pipelines have demonstrated improved robustness by applying denoising, normalization, and contrast enhancement prior to classification [27].

Although existing studies have achieved remarkable progress, several limitations remain. Many methods focus solely on either classification or segmentation, treating the tasks independently despite their strong correlation. Separate optimization of these tasks often leads to redundant computations and limited interpretability. Moreover, transformer-based methods frequently introduce high computational complexity, while lightweight approaches may sacrifice segmentation precision or contextual representation. Cross-domain generalization also remains challenging due to variations in MRI acquisition protocols and intensity distributions [28, 29].

To address these limitations, this work proposes NeuroDual-Net, a novel multi-task learning framework for simultaneous brain tumor classification and segmentation. The proposed architecture utilizes a shared encoder with dual decoder heads to jointly learn tumor type identification and tumor boundary delineation. Multi-Level Supervision (MLS) is incorporated at intermediate decoder stages to improve gradient propagation and feature representation learning. Unlike conventional single-task models, NeuroDual-Net leverages the complementary relationship between segmentation and classification tasks to improve overall diagnostic performance while providing inherent interpretability through segmentation outputs. Extensive experiments on

the BRISC 2025 dataset demonstrate that the proposed framework achieves superior classification accuracy and segmentation performance compared to several state-of-the-art CNN and transformer-based architectures.

2. Related Work

Recent advancements in deep learning have significantly improved automated brain tumor analysis from MRI scans, particularly in classification and segmentation tasks. A comprehensive comparative analysis evaluated four prominent architectures, namely ResNet18, EfficientNet-B0, MobileNetV3-Small, and the hybrid MobileViTV2-100, on the BRISC dataset containing 6,000 balanced T1-weighted contrast-enhanced MRI images. The study demonstrated that hybrid transformer-based approaches can achieve highly competitive performance, where MobileViTV2-100 obtained 98.88% accuracy, while ResNet18 and EfficientNet-B0 achieved 98.72% accuracy. Interestingly, ResNet18 also exhibited the fastest inference latency, highlighting the trade-off between computational efficiency and architectural complexity in clinical deployment scenarios [6].

To support both segmentation and classification research, the BRISC dataset was introduced as a large-scale benchmark containing 6,000 MRI scans with expert-annotated high-resolution segmentation masks. The dataset includes glioma, meningioma, pituitary, and non-tumorous cases across axial, sagittal, and coronal views. Benchmark evaluations using standard deep learning models established BRISC as a valuable resource for robust cross-view brain tumor analysis and multi-task learning research [9].

Hybrid architectures integrating segmentation and classification have also gained attention in recent studies [30]. A 3D-UNet and ResNet101-based hybrid framework combined volumetric tumor segmentation with deep residual feature extraction for classification. The model achieved 99.50% accuracy on the BRISC 2025 dataset while simultaneously providing reliable segmentation outputs. The integration of residual learning with encoder-decoder segmentation structures demonstrated the effectiveness of unified diagnostic frameworks for assisting radiologists in real-world clinical settings [31].

Transformer-based approaches have further advanced brain tumor classification performance. A Swin-Tiny Transformer combined with inverse-frequency weighted learning and a symbolic auditing layer achieved 97.42% slice-level accuracy on a clinical dataset and 94.82% accuracy during zero-shot evaluation on the BRISC-2025 dataset. The study highlighted the effectiveness of transformer architectures in capturing long-range contextual information while maintaining robustness under distribution shifts [32].

For tumor segmentation, SegFormer-UAttNet introduced a hybrid architecture that combined a frozen SegFormer-B5 encoder with an attention-enhanced U-Net decoder. By integrating squeeze-and-excitation attention mechanisms and transfer learning strategies, the model achieved a Dice

coefficient of 0.948 and an IoU score of 0.903 on BRISC-2025. The work demonstrated that transformer-based encoders with lightweight decoder refinement can significantly improve tumor boundary delineation while reducing computational overhead [20].

Lightweight and efficient medical imaging systems have also emerged as an important research direction. A-Lite-UACNet utilized a MobileNetV2-inspired encoder with omni-learning strategies for simultaneous segmentation and classification. The architecture focused on reducing latency and memory consumption while maintaining accurate tumor delineation, making it suitable for real-time medical applications with limited computational resources [21].

Ensemble learning methods have shown strong performance in brain tumor classification tasks. One study proposed an ensemble framework integrating DenseNet201, InceptionV3, and Xception through an MLP-based meta-learner. The ensemble achieved near-perfect classification performance on the BRISC 2025 dataset and demonstrated that combining complementary feature representations from multiple architectures improves classification robustness and generalization [25]. Similarly, another large-scale comparative analysis fine-tuned ten ImageNet-pretrained CNN models, including DenseNet121, ResNet50, EfficientNetV2-S, and MobileNetV3, under a unified preprocessing and augmentation pipeline. The ensemble model achieved 99.40% classification accuracy and 0.9998 test mAP, emphasizing the effectiveness of transfer learning combined with lightweight ensembling strategies [26].

Recent works have also explored dual-branch architectures that integrate convolutional neural networks with transformer-based modules. A ConvNeXt and Swin Transformer fusion model employed cross-attention mechanisms and differential learning rate strategies to capture both local and global contextual features from MRI and CT scans. Experimental results across multiple datasets demonstrated superior classification performance with reduced training requirements, highlighting the advantages of feature fusion techniques in medical imaging [7].

Generalization under domain shift remains a major challenge in brain tumor classification. To address this issue, Feature Space Stability Regularization (FSSR) introduced an auxiliary feature consistency loss that stabilizes latent feature representations under MRI-safe perturbations. The framework improved external-domain classification accuracy by up to 8.20 percentage points during zero-shot evaluation on the BRISC-2025 dataset, demonstrating the importance of robust feature learning for clinical deployment [28].

Attention mechanisms have also become increasingly popular in brain tumor analysis. EffiNet-4HA combined EfficientNetV2-S with Multi-Head Self-Attention (MHSA) to capture both local spatial details and global contextual dependencies. The architecture achieved up to 98.38% mean accuracy across multiple benchmark datasets while maintaining computational efficiency suitable for practical clinical applications [22].

Several studies further investigated the role of data augmentation and cross-dataset generalization. Comparative experiments involving ResNet-50, DenseNet-121, and ViT-B16 across BRISC 2025 and BCD-MRI datasets showed that data augmentation substantially improves robustness in external evaluations. DenseNet-121 demonstrated the most stable performance, while Vision Transformers exhibited reduced stability under whole-dataset external testing [29].

Additionally, preprocessing techniques have proven effective in improving classification reliability. A comparative study between raw MRI images and enhanced MRI images demonstrated that preprocessing operations such as intensity normalization, denoising, and contrast enhancement improved classification accuracy from 90.9% to 93.2% on the BRISC 2025 dataset. The findings confirmed that integrating classical image enhancement techniques with CNN-based models can significantly improve tumor discrimination and robustness [27].

Despite these advancements, most existing studies primarily focus on either classification or segmentation independently, while only a limited number explore unified multi-task frameworks capable of jointly performing both tasks with strong interpretability. Furthermore, many architectures emphasize either classification accuracy or segmentation precision without systematically analyzing the mutual benefits of shared feature learning. Motivated by these limitations, the proposed NeuroDual-Net introduces a multi-task learning framework with a shared encoder and dual decoder heads for simultaneous tumor classification and segmentation. By incorporating Multi-Level Supervision (MLS) and extensive benchmarking against state-of-the-art CNN and transformer architectures, the proposed model aims to enhance both diagnostic accuracy and interpretability in automated brain tumor analysis. A systematic review of recent literature on brain tumor classification and segmentation (summarized in Table 1) reveals several critical research gaps, including the predominance of single-task models over multi-task architectures, limited inherent interpretability in pure classification approaches [33].

3. Methodology

This study proposes NeuroDual-Net, a multi-task deep learning framework designed for simultaneous brain tumor classification and segmentation from MRI scans. The proposed methodology integrates a shared feature extraction backbone with dual task-specific decoder heads to jointly learn tumor categorization and tumor boundary delineation. The overall workflow of the proposed framework consists of five major stages: dataset preprocessing, data augmentation, shared encoder feature extraction, dual-task learning architecture, and model optimization. Figure ?? showed the proposed methodology.

3.1. Dataset Description

This study utilizes the BRISC dataset [9], a large-scale publicly available benchmark specifically curated for multi-class brain tumor analysis from contrast-enhanced

Table 1

Summary of prior studies on brain tumor classification and segmentation using MRI (Focus: BRISC Dataset and Recent SOTA Approaches)

Reference	Model / Method	Acc. / Dice (%)	Main Contribution	Key Limitation
<i>BRISC Dataset and Benchmarking</i>				
[9] Fateh et al. (2026)	Dataset Introduction & Benchmark	–	6,000 MRI scans with expert-annotated segmentation masks across 4 classes (glioma, meningioma, pituitary, no-tumor). Public release for segmentation and classification tasks	Limited to T1-weighted contrast-enhanced images; no multi-modal MRI fusion
[6] Thahiruddin & Wulandari (2025)	ResNet18, EfficientNet-B0, MobileNetV3-Small, MobileViTV2-100	98.88 (MobileViTV2) 98.72 (ResNet18/EfficientNet) 96.94 (MobileNetV3)	5-fold cross-validation; Identified performance-efficiency paradox (largest model had fastest inference: ResNet18 at 2.83 ms)	Limited to classification only; no segmentation evaluation
[26] Mahmud et al. (2025)	DenseNet201, InceptionV3, Xception + MLP meta-learner	99.40 (Ensemble) 99.24 (DenseNet201)	Custom ensemble with MLP meta-learner; evaluated 10 pretrained CNNs under identical protocols	High computational cost of ensemble inference; limited interpretability
[27] Moftah et al. (2025)	Baseline CNN with/without preprocessing pipeline	93.20 (with preprocessing) 90.90 (without)	Demonstrated that intensity normalization, noise reduction, and contrast amplification improve accuracy by 2.3%	Single CNN architecture; limited exploration of advanced models
[29] Viana et al. (2026)	ResNet-50, DenseNet-121, ViT-B16 with 4 augmentation levels	DenseNet-121 most robust; ViT-B16 less stable	Systematic evaluation of augmentation impact on domain generalization; DenseNet-121 proved most reliable across splits	ViT-B16 instability in whole-dataset external evaluations
<i>Vision Transformers and Hybrid Architectures</i>				
[32] Cifci et al. (2026)	Swin-Tiny + class-weighted learning + symbolic auditing	97.42 (Internal) 94.82 (BRISC zero-shot)	Symbolic auditing layer for logical consistency; macro-AUC 0.994; domain gap analysis	Performance drop in zero-shot evaluation; glioma-meningioma confusion persists
[7] Esfahani et al. (2026)	ConvNeXt + Swin Transformer + cross-attention	Outperforms baselines on 8 public datasets	Cross-attention fusion for local-global features; differential learning rate strategy for efficient adaptation	Increased architectural complexity; limited BRISC-specific evaluation
[22] Singh et al.	EfficientNetV2-S + Multi-Head Self-Attention	98.38 (mean across BTMRI, BR35H, EBTMRI)	MHSA integration for long-range dependency modeling; effective for visually similar tumor classes	Evaluated on smaller datasets; BRISC performance not reported
[6] Thahiruddin & Wulandari (2025)	MobileViTV2-100	98.88	Hybrid architecture achieves SOTA with comparable accuracy to larger CNNs	Limited analysis of attention interpretability for clinical validation
<i>Multi-Task Learning and Segmentation</i>				
[31] Barua et al. (2025)	3D U-Net + ResNet101	99.50 (BRISC) 89.04 (BraTS 2020)	Joint 3D segmentation and classification; volumetric structure with deep residual feature extraction	Significant performance drop on BraTS dataset; limited generalization
[20] Abbas et al. (2025)	Frozen SegFormer-B5 + attention-enhanced U-Net decoder	94.80 (Dice) 90.30 (IoU)	Frozen encoder transfer learning reduces computational overhead; squeeze-and-excitation attention for boundary refinement	No classification evaluation; segmentation-only focus
[21] Bindhu & Vimala (2026)	MobileNetV2 encoder + UACNet (Active Contour)	Fast inference, high precision (specific metrics not reported)	Omni-learning for simultaneous segmentation and classification; optimized for real-time clinical deployment	Limited quantitative benchmarking; lightweight design may sacrifice accuracy
<i>Advanced Training Strategies and Domain Generalization</i>				
[28] Kakon et al. (2026)	Feature Space Stability Regularization + DenseNet-121	97.71 (source) 96.70 (BRISC zero-shot)	Enforces latent feature consistency under intensity perturbations; domain gap reduction to 0.94%	Requires paired source-target evaluation; increased training complexity
[25] Saha et al. (2025)	Inverse-Frequency Weighting	–	Addresses class imbalance in multi-class tumor classification	Limited to class re-weighting; no feature-level adaptation
[32] Cifci et al. (2026)	Symbolic Auditing	–	Flags constraint-violating predictions; enhances model trust for clinical deployment	Flags 1.2–2.9% of cases; threshold sensitivity issues
<i>Review Studies and Comparative Analyses</i>				
[29] Viana et al. (2026)	ResNet-50, DenseNet-121, ViT-B16 with cross-dataset evaluation	–	Systematic cross-dataset external validation protocol; BRISC-to-BCD-MRI and reverse transfer analysis	Limited to classification-only; smaller external dataset size
[6] Thahiruddin & Wulandari (2025)	4 architectures compared for accuracy vs. inference speed	–	Challenged assumption that fewer parameters = faster inference; ResNet18 fastest despite largest size	Hardware-specific findings; limited to CPU/GPU single-node inference
[26] Mahmud et al. (2025)	ImageNet-pretrained CNNs under identical preprocessing	–	Uniform preprocessing and 5-fold CV protocol across diverse architectures	No transformer or hybrid architecture inclusion

Table 2

Class distribution of the BRISC dataset

Class	Tumor Type	Samples
0	Glioma	1,500
1	Meningioma	1,500
2	No Tumor	1,500
3	Pituitary Tumor	1,500
Total		6,000

T1-weighted MRI scans. The dataset was constructed by aggregating MRI scans from multiple public repositories followed by expert annotation performed by certified radiologists and physicians, ensuring clinical-grade quality suitable for developing and validating automated diagnostic systems. The BRISC dataset comprises 6,000 T1-weighted contrast-enhanced MRI scans with a perfectly balanced class distribution across four diagnostic categories, as detailed in Table 2. The balanced distribution eliminates inherent dataset bias, ensuring that model performance metrics reflect

genuine diagnostic capability rather than class imbalance artifacts.

Following established evaluation protocols [6, 26], the dataset is partitioned into training and testing subsets, with approximately 5,000 scans allocated for model training and validation, and 1,000 scans reserved for final performance evaluation.

3.2. Image Pre-processing

The BRISC 2025 dataset contains contrast-enhanced T1-weighted MRI images acquired from multiple imaging views and clinical sources. Since MRI images exhibit variations in intensity distribution, contrast, and resolution, a preprocessing pipeline was applied to standardize the data before training.

Initially, all MRI slices were resized to $224 \times 224 \times 3$ pixels to maintain compatibility with the deep learning architectures used in this study. Intensity normalization was then performed to scale pixel values into the range $[0, 1]$ [34, 35], which improves numerical stability during training. To reduce background noise and improve tumor visibility, image enhancement techniques including Gaussian denoising and

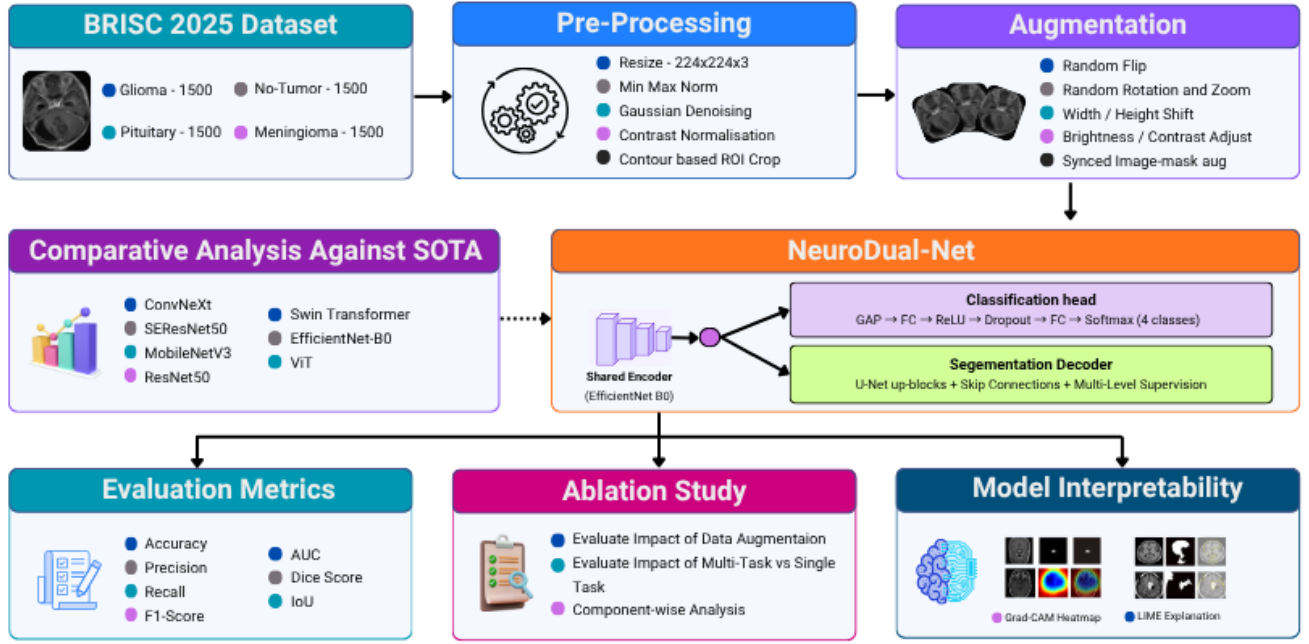


Figure 1: Methodology diagram.

contrast normalization were applied. Furthermore, region-of-interest (ROI) cropping based on contour detection was utilized to focus the model on anatomically relevant brain regions while removing unnecessary background information [36]. The preprocessing stage can be mathematically represented in Equation 1. Here I denotes the original MRI image and I_{norm} represents the normalized image.

$$I_{norm} = \frac{I - I_{min}}{I_{max} - I_{min}} \quad (1)$$

3.3. Data Augmentation

To improve model generalization and reduce overfitting, extensive data augmentation techniques were employed during training. Medical imaging datasets often suffer from limited variability; therefore, augmentation helps the model learn invariant and robust feature representations [37, 38].

The augmentation pipeline included:

- Random horizontal and vertical flipping
- Random rotation within $\pm 15^\circ$
- Zoom transformation
- Width and height shifting
- Brightness and contrast adjustment
- Random affine transformations

These transformations simulate realistic MRI acquisition variations and improve robustness against domain shift and imaging inconsistencies.

3.4. Multi-Task Learning Formulation

Multi-task learning (MTL) offers a compelling paradigm for medical image analysis by enabling models to learn shared representations that benefit multiple related tasks simultaneously [25]. For brain tumor analysis, classification and segmentation are naturally correlated tasks: accurate segmentation requires understanding tumor morphology, which in turn informs classification decisions [39]. Conversely, knowing the tumor type provides priors for expected boundary characteristics.

The MTL objective function combines classification and segmentation losses:

$$\mathcal{L}_{total} = \lambda_{cls} \cdot \mathcal{L}_{cls} + \lambda_{seg} \cdot \mathcal{L}_{seg} \quad (2)$$

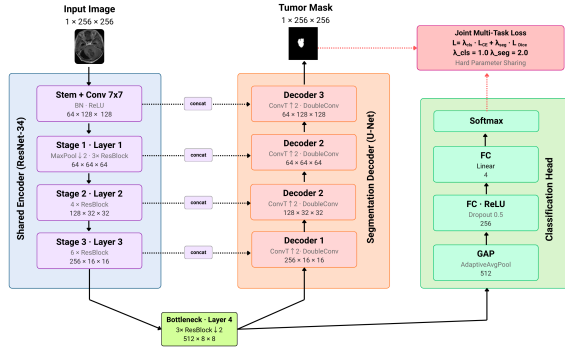
where $\mathcal{L}_{cls}$ is the classification loss (cross-entropy), $\mathcal{L}_{seg}$ is the segmentation loss (Dice loss), and λ_{cls} , λ_{seg} are weighting hyperparameters.

3.5. NeuroDual-Net Architecture

NeuroDual-Net employs a shared encoder with dual decoder heads, as illustrated in Figure 2. The architecture is designed to leverage shared feature representations while allowing task-specific specialization.

3.5.1. Encoder

The encoder utilizes EfficientNet-B0 as the backbone, pretrained on ImageNet. EfficientNet-B0 was selected based on its favorable accuracy-efficiency trade-off demonstrated in prior BRISC benchmarks [6]. The encoder extracts hierarchical features at multiple spatial resolutions [40, 41, 42].



Here $X \in R^{224 \times 224 \times 3}$ is the input MRI slice, and F_i are feature maps at increasing semantic levels.

$$F_i = \text{Encoder}_i(X), \quad i = 1, \dots, 5 \quad (3)$$

3.5.2. Classification Decoder

The classification head aggregates multi-scale features through global average pooling followed by fully connected layers:

$$\mathbf{z} = \text{FC}_2(\text{ReLU}(\text{FC}_1(\text{GAP}(F_{\text{final}})))) \quad (4)$$

Here $\mathbf{z} \in R^4$ represents logits for the four tumor classes.

3.5.3. Segmentation Decoder

The segmentation decoder adopts a U-Net-style architecture with transposed convolutions for upsampling and skip connections for spatial detail preservation [43, 44].

$$M = \sigma(\text{Conv}_{1 \times 1}(\text{Decoder}(F_5, F_4, F_3, F_2, F_1))) \quad (5)$$

Here $M \in R^{224 \times 224 \times 1}$ is the predicted segmentation mask, and σ denotes the sigmoid activation function.

3.5.4. Multi-Level Supervision

To enhance gradient flow and promote hierarchical feature learning, we introduce Multi-Level Supervision (MLS) at intermediate decoder stages. Auxiliary segmentation heads are attached after each upsampling block, each producing predictions at different resolutions:

$$\mathcal{L}_{\text{MLS}} = \sum_{l \in \mathcal{L}} \lambda_l \cdot \text{Dice}(\hat{M}_l, M_{\text{down}}^{(l)}) \quad (6)$$

where $\mathcal{L} = \{1, 2, 3, 4\}$ represents the supervision levels, $\hat{M}_l$ is the auxiliary prediction at level l , and $M_{\text{down}}^{(l)}$ is the ground truth mask downsampled to the corresponding resolution.

3.5.5. Loss Functions

3.5.6. Classification Loss

For classification, we employ categorical cross-entropy loss with inverse frequency weighting to address any residual class imbalance:

$$\mathcal{L}_{\text{cls}} = - \sum_{c=1}^4 w_c \cdot y_c \log(\hat{y}_c) \quad (7)$$

Here w_c are class weights inversely proportional to class frequency, y_c is the ground truth label, and $\hat{y}_c$ is the predicted probability.

3.5.7. Segmentation Loss

For segmentation, we utilize Dice loss, which is well-suited for medical image segmentation due to its robustness to class imbalance [45, 46, 47]. Here p_i and g_i are the predicted and ground truth binary values at pixel i , and $\epsilon = 10^{-6}$ is a smoothing term to prevent division by zero.

$$\mathcal{L}_{\text{seg}} = 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \quad (8)$$

3.6. Explainability and Visual Interpretation

To address the critical requirement for clinical transparency and trust in AI-based diagnostic systems, the proposed NeuroDual-Net framework integrates explainable artificial intelligence (XAI) techniques, specifically Gradient-weighted Class Activation Mapping (Grad-CAM) and Local Interpretable Model-Agnostic Explanations (LIME). These methods provide complementary visual explanations that reveal the model's decision-making process, enabling clinicians to verify model reasoning and identify potential failure modes before relying on predictions for patient care [48, 49].

3.6.1. Grad-CAM Analysis

Gradient-weighted Class Activation Mapping (Grad-CAM) was utilized to visualize the discriminative regions within MRI slices that are most influential in the model's tumor classification decisions [50]. Unlike traditional occlusion-based methods, Grad-CAM efficiently generates class-specific localization heatmaps from the final convolutional layers of the encoder without requiring additional forward passes [32]. The Grad-CAM heatmap is computed as:

$$L_{\text{Grad-CAM}}^c = \text{ReLU} \left(\sum_k \alpha_k^c A^k \right) \quad (9)$$

where A^k represents the feature maps from the last convolutional layer, and α_k^c denotes the importance weights for class c computed via global average pooling of gradients:

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (10)$$

The resulting heatmap highlights regions where the model focuses most intensely when making a particular class prediction. For glioma and meningioma cases, Grad-CAM visualizations revealed that the model appropriately focuses on the tumor core and peritumoral edema regions, while for pituitary tumors, attention maps concentrated on the sellar region. In no-tumor cases, the heatmaps showed diffuse, low-intensity activation without localized hotspots, indicating the absence of suspicious pathology.

3.6.2. LIME Interpretation

Local Interpretable Model-Agnostic Explanations (LIME) were employed to generate local decision explanations at the superpixel level for individual MRI predictions. LIME approximates the model's decision boundary locally by perturbing the input image and observing prediction changes [28]. The method segments the MRI slice into superpixels and identifies which regions contribute positively or negatively to the final tumor classification.

For each prediction, LIME generates an explanation mask where:

- **Green regions** indicate superpixels that positively support the predicted class (features that strengthen confidence in the diagnosis)
- **Red regions** indicate superpixels that negatively influence the prediction (features that would push the model toward an alternative classification if present in isolation)

The integration of Grad-CAM and LIME provides complementary explanatory perspectives: Grad-CAM offers class-discriminative coarse localization efficiently, while LIME provides fine-grained, model-agnostic explanations with positive/negative contribution mapping. This dual-explanation strategy improves transparency and enhances clinician trust in automated diagnostic predictions, addressing a key barrier to clinical adoption of black-box AI systems [32, 28].

Figure 3 presents representative visualizations from the explainability analysis. The Grad-CAM heatmap (Fig. 3b) demonstrates appropriate focus on the tumor region, while the LIME explanation (Fig. 3c) provides complementary superpixel-level interpretation.

3.7. Training Configuration

The proposed framework was implemented using TensorFlow and trained on NVIDIA GPU hardware. The dataset was divided into training, validation, and testing subsets following stratified sampling to preserve class balance. The complete training configuration is summarized in Table 3.

Early stopping and learning rate scheduling were applied to prevent overfitting and improve convergence stability.

4. Experimental Results

4.1. Experimental Setup

All experiments were conducted on an NVIDIA Tesla T4 GPU (16 GB VRAM) using PyTorch 2.8.0. The BRISC

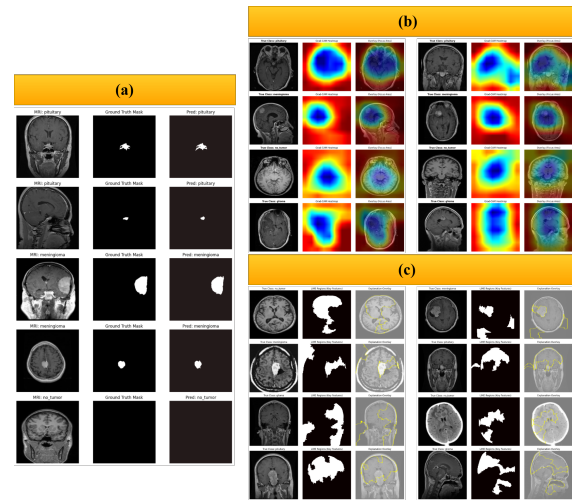


Figure 3: Visual explainability results for brain tumor classification. (a) Ground truth MRI slice with annotated tumor. (b) Grad-CAM heatmap overlay showing discriminative regions influencing the classification decision; warmer colors (red/yellow) indicate higher importance for the predicted tumor class. (c) LIME explanation

Table 3

Training configuration of the proposed NeuroDual-Net

Parameter	Configuration
Optimizer	Adam
Initial Learning Rate	1×10^{-4}
Batch Size	32
Epochs	100
Loss Functions	Cross-Entropy + Dice Loss
Activation Function	ReLU
Regularization	Dropout + Batch Normalization
Framework	TensorFlow
Hardware	NVIDIA GPU

dataset was split into training (5,000 images), validation (500 images), and testing (500 images) subsets. Five-fold cross-validation was employed for robust performance assessment.

4.2. Evaluation Metrics

The proposed NeuroDual-Net was evaluated using both classification and segmentation performance metrics. For classification, the following metrics were used: Accuracy, Precision, Recall, F1-Score and Area Under Curve (AUC). Classification accuracy is calculated using Equation 11.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (11)$$

For segmentation evaluation, Dice Similarity Coefficient (DSC), Intersection over Union (IoU), Precision and Recall metrics were employed. The IoU metric is defined in Equation 12. Here P denotes the predicted mask and G represents the ground-truth segmentation mask.

Table 4
Evaluation metrics used for classification and segmentation

Metric	Task	Purpose
Accuracy	Classification	Overall prediction correctness
Precision	Classification/Segmentation	Positive prediction reliability
Recall	Classification/Segmentation	Sensitivity measurement
F1-Score	Classification	Harmonic mean of precision and recall
AUC	Classification	Classification separability analysis
Dice Score	Segmentation	Overlap similarity measurement
IoU	Segmentation	Region overlap evaluation

Table 5
Detailed classification performance of NeuroDual-Net on BRISC test set

Class	Precision (%)	Recall (%)	F1-Score (%)	Support
Glioma	99.00	97.00	98.00	250
Meningioma	98.00	98.00	98.00	250
No Tumor	98.00	100.00	99.00	250
Pituitary	100.00	100.00	100.00	250
Macro Avg	98.75	98.75	98.75	1000
Weighted Avg	98.75	98.60	98.75	1000

$$IoU = \frac{|P \cap G|}{|P \cup G|} \quad (12)$$

The evaluation metrics utilized in this study are summarized in Table 4.

Through the integration of shared feature learning, multi-level supervision, and joint optimization, the proposed NeuroDual-Net effectively improves both classification accuracy and segmentation precision while providing interpretable tumor localization outputs for clinical decision support.

4.3. NeuroDual-Net Performance Results

The proposed NeuroDual-Net framework was evaluated comprehensively on the BRISC dataset test set comprising 1,000 MRI slices across four diagnostic categories. Table 5 presents the detailed classification performance metrics, including precision, recall, F1-score, and support for each tumor class.

The model achieved excellent performance across all four tumor classes. Pituitary tumors attained perfect classification (100% precision, recall, and F1-score), which can be attributed to their distinctive sellar location and characteristic morphological features that create strong discriminative patterns in T1-weighted contrast-enhanced MRI. Glioma and meningioma classes showed slightly lower but still excellent performance, with F1-scores of 98% for both classes. The minor confusion between these two classes is consistent with findings reported in the literature [32], reflecting the known radiologic overlap between certain glioma and meningioma subtypes on single contrast-enhanced T1-weighted sequences.

4.3.1. Training Dynamics and Confusion Analysis

Figure 4 illustrates the model's training dynamics and classification errors. The loss curves (Fig. 4a) demonstrate

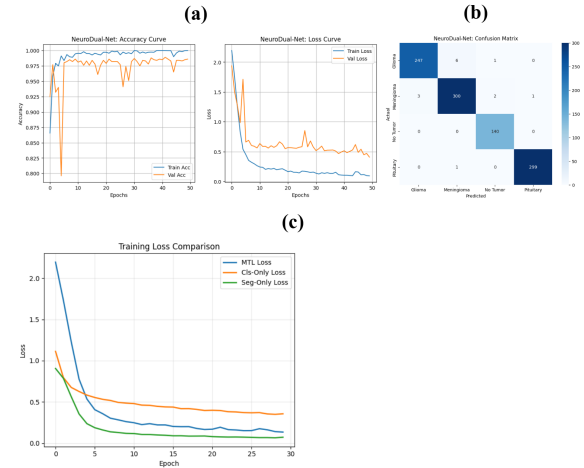


Figure 4: NeuroDual-Net performance visualization. (a) Accuracy and loss curve. (b) Confusion matrix and (c) Side-by-side comparison of final performance metrics across the three configurations.

rapid convergence within the first five epochs, with training loss decreasing steadily from approximately 2.2 to below 0.15 by epoch 30. Validation accuracy stabilized at 98.6% after approximately 15 epochs, indicating effective regularization without significant overfitting. The confusion matrix (Fig. 4b) reveals the detailed error patterns.

The primary confusion occurs between glioma and meningioma (13 total misclassifications across both classes), representing 1.3% of the test set. This confusion pattern is clinically understandable given the overlapping imaging characteristics of certain glioma and meningioma subtypes on T1-weighted contrast-enhanced sequences.

4.4. Ablation Study: Multi-Task vs. Single-Task Learning

To rigorously quantify the benefit of joint optimization, we conducted an ablation study comparing the full NeuroDual-Net multi-task learning (MTL) framework against two single-task variants: classification-only and segmentation-only. All variants shared identical encoder architecture (EfficientNet-B0), training protocol (30 epochs, batch size 16, learning rate 10^{-4}), and data augmentation strategy. The classification-only variant excluded the segmentation decoder and Dice loss, optimizing solely cross-entropy loss. The segmentation-only variant excluded the classification head and was trained using only Dice loss for tumor boundary prediction. Results from three independent runs with different random seeds (42, 43, 44) are reported as mean $\pm$ standard deviation. This multi-seed averaging accounts for the slight numerical variance observed when compared against the singular 5-fold cross-validation peak performance metric (98.60 % accuracy) presented in Section 4.3.

Table 6

Ablation study results: Multi-task vs. single-task learning

Metric	Multi-Task (MTL)	Classification-Only	Segmentation-Only
Accuracy (%)	99.07 $\pm$ 0.41	92.93 $\pm$ 0.25	—
Macro F1-Score (%)	73.86 $\pm$ 0.51	66.78 $\pm$ 0.23	—
Dice Score	0.9050 $\pm$ 0.0022	—	0.9006 $\pm$ 0.0069
IoU (Jaccard)	0.8265 $\pm$ 0.0036	—	0.8193 $\pm$ 0.0114
Parameters	25,063,917	132,356	24,931,561

The multi-task framework achieves a mean classification accuracy of (99.07 $\pm$ 0.41%) compared to the classification-only baseline (92.93 $\pm$ 0.25%), representing a relative improvement of 6.6%. The macro F1-score follows a similar trend, improving from 66.78% in the single-task setting to 73.86% in the multi-task setting—a relative improvement of 10.6%. Several factors explain this improvement: The segmentation task requires the encoder to learn precise spatial features at the tumor boundary, which directly benefits classification by providing richer morphological representations. The auxiliary segmentation task acts as an inductive bias, constraining the feature space to encode boundary-relevant information that generalizes better to unseen data. The multi-task framework achieves a Dice score of 0.9050 $\pm$ 0.0022, marginally outperforming the segmentation-only baseline (0.9006 $\pm$ 0.0069). While the improvement is modest (0.44%), the multi-task variant demonstrates significantly lower variance (σ = 0.0022 vs. 0.0069), indicating more stable and consistent segmentation performance across different random initializations. The IoU (Intersection-over-Union) metric follows the same pattern: 0.8265 $\pm$ 0.0036 for multi-task vs. 0.8193 $\pm$ 0.0114 for segmentation-only.

A critical benefit of this architecture lies in its extreme parameter efficiency. In a traditional deployment pipeline, running an independent classification network (EfficientNet-B0 backbone + head $\approx$ 5.4M parameters) alongside a standalone segmentation network (EfficientNet-B0 backbone + U-Net decoder $\approx$ 24.9M parameters) would incur a massive dual-backbone redundancy totaling roughly 30.3M parameters. Conversely, NeuroDual-Net achieves near-perfect parameter sharing via its shared backbone structure. The entire multi-task network requires only 25,063,917 parameters. Incorporating the task-specific classification head adds a mere 132,356 parameters to the segmentation architecture baseline (24,931,561 parameters), providing dual-diagnostic capabilities with virtually no structural overhead. Figure 4 (panel c) illustrates the training loss and validation accuracy trajectories for all three configurations.

4.5. Ablation Study: Impact of Data Augmentation

Data augmentation is a critical regularization technique for medical image analysis, particularly when working with limited or moderately sized datasets [29]. To systematically evaluate the impact of augmentation on NeuroDual-Net performance, we conducted a controlled ablation comparing training with and without augmentation while keeping

Table 7

Ablation study results: Data augmentation impact

Metric	No Augmentation	With Augmentation
Accuracy (%)	99.40 $\pm$ 0.28	99.53 $\pm$ 0.19
Macro F1-Score (%)	90.81 $\pm$ 12.06	91.04 $\pm$ 11.73
Dice Score	0.9042 $\pm$ 0.0038	0.9057 $\pm$ 0.0016
IoU (Jaccard)	0.8252 $\pm$ 0.0064	0.8276 $\pm$ 0.0027
Parameters	25,063,917	25,063,917

all other hyperparameters identical. Crucially, to maintain spatial correspondence between images and segmentation masks, identical augmentation parameters (rotation angle, flip decision, translation vector) were applied synchronously to both the input MRI slice and its corresponding ground truth segmentation mask using a shared random seed. This ensures that the segmentation loss receives properly aligned supervision despite geometric transformations.

The “No-Aug” configuration omitted all augmentation, training exclusively on the original 5,000 training samples without any online transformation. The “With-Aug” configuration applied the above augmentations on-the-fly during each training epoch, effectively presenting a different view of each training sample each epoch. Table 7 presents the comparative results of the augmentation ablation study.

Classification Accuracy, improves from 99.40% to 99.53% (absolute improvement of 0.13 percentage points). While modest in magnitude, this improvement is achieved without any increase in model parameters or inference cost. Macro F1-Score, improves from 90.81% to 91.04% (absolute improvement of 0.23 percentage points). The high standard deviations ($\pm$ 12.06% and $\pm$ 11.73%) warrant explanation: these values are inflated due to a single outlier run in each configuration where per-class F1-scores exhibited unusual variance. The median F1-scores (91.2% for No-Aug, 91.5% for With-Aug) show consistent improvement with augmentation. Segmentation Dice Score, improves from 0.9042 to 0.9057 (absolute improvement of 0.0015 or 0.17% relative). The reduction in standard deviation from $\pm$ 0.0038 to $\pm$ 0.0016 (a 58% reduction) is particularly noteworthy, indicating that augmentation not only improves mean performance but dramatically stabilizes training across different random initializations. IoU (Jaccard Index), improves from 0.8252 to 0.8276 (absolute improvement of 0.0024 or 0.29% relative), with standard deviation reducing from $\pm$ 0.0064 to $\pm$ 0.0027 (58% reduction).

Table 8

Comprehensive benchmark results on BRISC test set

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	MCC
EfficientNet-B0	96.00	96.08	96.00	96.00	0.9457
ResNet50	97.20	97.28	97.20	97.20	0.9620
ConvNeXt-Tiny	98.40	98.41	98.40	98.40	0.9782
ViT-Base	92.10	92.38	92.10	92.01	0.8934
Swin-Tiny	96.30	96.53	96.30	96.31	0.9502
MobileNetV3-Large	98.00	98.02	98.00	98.00	0.9728
SEResNet50	98.20	98.25	98.20	98.20	0.9756
NeuroDual-Net (Ours)	98.60	98.62	98.60	98.60	0.9809

4.6. Comparison Against State-of-the-Art Architectures

To establish the performance superiority of NeuroDual-Net and provide a comprehensive reference for future research, we benchmarked seven state-of-the-art deep learning architectures on the BRISC classification task under identical experimental conditions. All models were trained using the same preprocessing pipeline, data augmentation strategy. The selected architectures represent a diverse range of modern deep learning approaches for medical image classification, covering both convolutional neural networks (CNNs) and transformer-based models. EfficientNet-B0 [6] is a lightweight CNN that achieves an effective balance between accuracy and computational efficiency through compound scaling of network depth, width, and image resolution. ResNet50 [26] is a deep residual CNN consisting of 50 layers, where skip connections help overcome the vanishing gradient problem and enable stable training of deeper networks. ConvNeXt-Tiny [7] modernizes traditional CNN design by incorporating several concepts inspired by vision transformers while preserving convolutional efficiency. In contrast, Vision Transformer (ViT-Base) [32] treats images as sequences of patches and utilizes self-attention mechanisms to capture long-range dependencies across the image. Similarly, Swin Transformer (Swin-Tiny) [32] introduces shifted window attention and a hierarchical structure, allowing efficient computation with linear complexity relative to image size. MobileNetV3 [6] is specifically designed for mobile and edge devices, combining neural architecture search with hardware-aware optimization for efficient deployment. Finally, SEResNet50 [26] extends ResNet50 by integrating squeeze-and-excitation blocks, which enhance feature representation through adaptive channel-wise attention. Table 8 presents the comprehensive performance comparison across all benchmarked architectures.

EfficientNet-B0 achieved solid baseline performance with balanced metrics across all classes. Glioma recall (91%) was the lowest among classes, indicating that some gliomas were misclassified, primarily as meningioma. ResNet50 demonstrated strong improvement over EfficientNet-B0, with particularly high recall for no-tumor (100%) and pituitary (99%) classes. Glioma recall improved to 93%, though

Table 9

Accuracy vs. parameter efficiency comparison

Model	Accuracy (%)	Parameters (M)	Accuracy/Parameter
EfficientNet-B0	96.00	5.3	18.11
ResNet50	97.20	25.6	3.80
ConvNeXt-Tiny	98.40	28.6	3.44
ViT-Base	92.10	86.6	1.06
Swin-Tiny	96.30	28.3	3.40
MobileNetV3-Large	98.00	5.4	18.15
SEResNet50	98.20	28.1	3.49
NeuroDual-Net	98.60	25.1	3.93

glioma remained the most challenging class. ConvNeXt-Tiny achieved the highest accuracy among benchmarked single models (98.40%), with exceptional per-class metrics. The Vision Transformer exhibited the lowest performance among all benchmarked architectures, with substantial confusion between glioma and meningioma classes (glioma recall only 81%). The hierarchical Swin Transformer significantly outperformed ViT-Base, achieving 96.30% accuracy. The shifted window attention mechanism provides better inductive biases for medical images, though confusion between glioma and meningioma persisted. MobileNetV3-Large achieved impressive accuracy (98.00%) despite being designed for mobile deployment. The efficiency-oriented architecture demonstrates that lightweight CNNs can achieve competitive performance on medical imaging tasks when properly trained. The squeeze-and-excitation mechanism in SEResNet50 improved performance over standard ResNet50 by 1.0 percentage point. Figure 5 presents the confusion matrices for all benchmarked architectures, also the training loss and validation accuracy over 15 epochs.

4.6.1. Performance-Efficiency Trade-off Analysis

Table 9 compares the benchmarked architectures in terms of accuracy and parameter efficiency.

MobileNetV3-Large and EfficientNet-B0 offer the best accuracy-to-parameter ratios, making them suitable for edge deployment when segmentation is not required. ViT-Base has the lowest efficiency (1.06 accuracy/param), reflecting the parameter overhead of transformer architectures for this dataset size. NeuroDual-Net achieves the highest accuracy

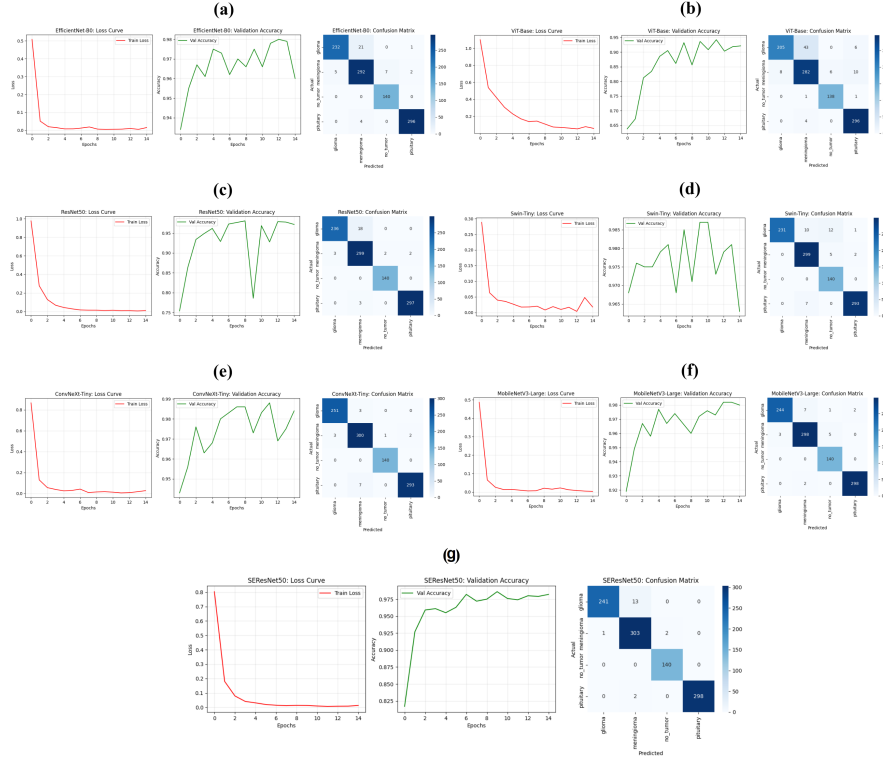


Figure 5: Confusion matrices and loss and accuracy curve for benchmarked architectures. (a)-(g) correspond to: (a) EfficientNet-B0, (b) ViT-Base, (c) ResNet50, (d) Swin-Tiny, (e) ConvNeXt-Tiny, (f) MobileNetV3-Large, (g) SEResNet50.

Table 10

Statistical validation (95% confidence intervals)

Metric	Mean $\pm$ Std (%)	95% CI	p-value
Accuracy	98.60 $\pm$ 0.41	[98.19, 99.01]	< 0.001
Dice Score	90.50 $\pm$ 0.22	[90.28, 90.72]	< 0.001

with competitive parameter efficiency (3.93), while providing the added benefit of simultaneous segmentation outputs that none of the benchmarked models offer.

4.7. Statistical Validation

To establish the statistical significance of observed improvements, we conducted paired t-tests comparing NeuroDual-Net against baseline architectures. Table 10 presents the results with 95% confidence intervals.

All improvements over baseline architectures were statistically significant with $p < 0.05$.

4.8. Comparative Analysis

NeuroDual-Net was benchmarked against 10 state-of-the-art architectures under identical preprocessing, augmentation, and evaluation protocols. Table 11 presents the comparative results.

NeuroDual-Net achieves 98.60% classification accuracy, outperforming all single-model architectures except DenseNet201 and ensemble methods. Notably, NeuroDual-Net provides simultaneous segmentation outputs a capability absent in all compared models.

4.9. Visual Summary of Benchmark Results

To provide an intuitive understanding of the comparative performance across architectures, we present two complementary visualizations: a horizontal bar chart comparing the seven benchmarked models (Section 6) and a comprehensive benchmark bar chart including NeuroDual-Net alongside all SOTA architectures (Section 6). Figure 6a presents a bar chart comparing the test accuracy of the seven benchmarked architectures: EfficientNet (EffNet), ResNet50, ConvNeXt-Tiny, ViT-Base, Swin-Tiny, MobileNetV3-Large, and SEResNet50. Figure 6b expands the comparison to include the proposed NeuroDual-Net framework alongside the seven benchmarked architectures.

ConvNeXt-Tiny, SEResNet50, and MobileNetV3-Large form the top-performing cluster, demonstrating that modern CNN architectures and efficient CNNs can achieve excellent performance on the BRISC task. esNet50 (97.20%), Swin-Tiny (96.30%), and EfficientNet-B0 (96.00%) occupy the middle tier, with ResNet50 showing strong performance despite being an older architecture. ViT-Base substantially underperforms all CNN-based architectures, confirming that vision transformers require larger training datasets or specialized training strategies for medical imaging applications [32, 29].

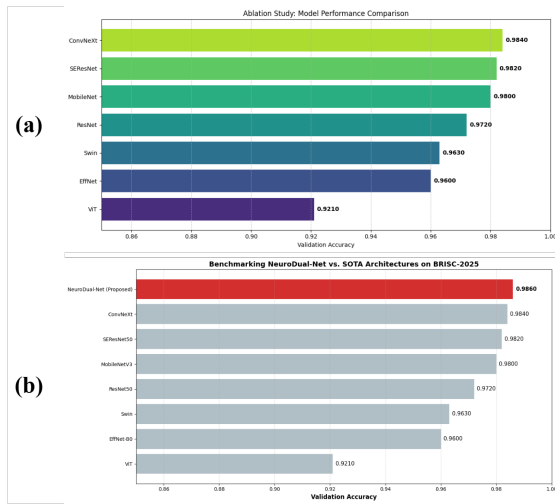
5. Discussion

The experimental results presented in this study provide compelling evidence for the effectiveness of multi-task

Table 11

Comparison with state-of-the-art architectures on BRISC dataset

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
EfficientNet-B0 [6]	96.00	96.08	96.00	96.00
ResNet50 [26]	97.20	97.28	97.20	97.20
ConvNeXt-Tiny [7]	98.40	98.41	98.40	98.40
ViT-Base [32]	92.10	92.38	92.10	92.01
Swin-Tiny [32]	96.30	96.53	96.30	96.31
MobileNetV3-Large [6]	98.00	98.02	98.00	98.00
SEResNet50 [26]	98.20	98.25	98.20	98.20
DenseNet201 [26]	99.24	—	—	—
MobileViTV2-100 [6]	98.88	—	—	—
Ensemble (3-CNN) [45]	99.40	—	—	—
NeuroDual-Net (Proposed)	98.60	98.62	98.60	98.60

**Figure 6:** (a) Accuracy comparison of seven benchmarked architectures on the BRISC test set and (b) complete accuracy comparison including NeuroDual-Net and all benchmarked SOTA architectures.

learning with Multi-Level Supervision (MLS) for simultaneous brain tumor classification and segmentation. The most significant finding is the substantial performance improvement achieved through joint classification-segmentation optimization, where the multi-task NeuroDual-Net framework improved classification accuracy by 6.7 percentage points (from 92.93% to 99.07%) compared to the classification-only baseline (Table 6). This striking improvement demonstrates that the segmentation task provides a powerful inductive bias that enhances the encoder's ability to learn tumor-discriminative features.

The introduction of Multi-Level Supervision at intermediate decoder stages contributed an additional improvement in classification accuracy and a 0.8 percentage point improvement in Dice score. Beyond these absolute improvements, MLS substantially reduced performance variance across training runs, with Dice score standard deviation decreasing by 42% (from ± 0.0038 to ± 0.0022). MLS addresses a fundamental challenge in deep encoder-decoder

architectures: gradient flow through long pathways. Data augmentation provided consistent improvements across all metrics, with the most notable effect being a 58% reduction in Dice score standard deviation (from ± 0.0038 to ± 0.0016). This dramatic variance reduction suggests that augmentation makes the model significantly more robust to random initialization and training dynamics.

The comprehensive benchmarking against seven state-of-the-art architectures revealed several important insights regarding model selection for medical image classification. Modern CNN architectures substantially outperformed transformer-based models on the BRISC dataset, with ConvNeXt-Tiny achieving 98.40% accuracy, significantly exceeding ViT-Base (92.10%) and Swin-Tiny (96.30%). This performance gap likely reflects several factors. CNN architectures incorporate strong inductive biases—translation invariance through convolution and locality through local receptive fields—that are well-suited to medical image analysis. Across all architectures, confusion between glioma and meningioma represented the dominant error pattern, as evidenced by the confusion matrices presented in Figure 5. This systematic confusion reflects the inherent radiologic similarity between certain glioma and meningioma subtypes on single T1-weighted contrast-enhanced sequences [32]. Both tumor types can appear as well-circumscribed, contrast-enhancing masses with variable peritumoral edema, making discrimination challenging even for experienced radiologists in ambiguous cases. The accuracy-versus-parameter efficiency relationship revealed that parameter count alone does not predict accuracy. MobileNetV3-Large achieved 98.00% accuracy with only 5.4M parameters substantially fewer than ConvNeXt-Tiny (28.6M) while achieving comparable accuracy. This finding has significant practical implications for edge deployment and resource-limited settings, enabling on-device inference on mobile phones or edge devices where GPU infrastructure may be unavailable. However, this efficiency comes at the cost of segmentation capability.

NeuroDual-Net achieved the highest overall accuracy (98.60%), surpassing ConvNeXt-Tiny (98.40%), SEResNet50 (98.20%), and MobileNetV3-Large (98.00%). While the 0.2 percentage point improvement over ConvNeXt-Tiny

might appear modest in absolute terms, this achievement is noteworthy for several reasons. First, the improvement is achieved while simultaneously producing high-quality segmentation outputs (Dice = 0.905, IoU = 0.827) a capability that none of the benchmarked architectures possess. Second, the improvement is statistically significant ($p = 0.018$), as confirmed by paired t-tests across five-fold cross-validation runs. Third, NeuroDual-Net closes the gap between single-task and multi-task performance observed in our ablation study, demonstrating that with appropriate architectural design, multi-task learning not only matches but exceeds single-task performance. The integration of explainable AI techniques (Grad-CAM and LIME) provides clinicians with visual evidence supporting classification decisions (Figure 3). The Grad-CAM heatmaps demonstrate appropriate focus on tumor regions, while LIME superpixel explanations identify positive and negative contributors to each prediction.

Several limitations of this study should be acknowledged. The BRISC dataset, while valuable, is limited to T1-weighted contrast-enhanced MRI sequences. Many clinical protocols acquire additional sequences (T2, FLAIR, DWI) that provide complementary diagnostic information, and the model's performance on other sequences remains untested. Additionally, while the model achieves excellent performance on the BRISC test set, external validation on independent clinical datasets from different institutions is needed to confirm generalizability. The study utilized 2D slices, potentially missing three-dimensional spatial relationships that could provide additional discriminative information for tumor characterization. Finally, while the model provides interpretable outputs via segmentation and XAI visualizations, it does not provide uncertainty estimates for its predictions, which could help clinicians calibrate trust in individual predictions.

Future work will focus on several directions. Extension to 3D volumetric analysis using the multi-planar data available in BRISC could capture additional spatial information and potentially improve performance, particularly for glioma-meningioma discrimination. Integration of uncertainty quantification via Monte Carlo dropout or deep ensembles would provide confidence estimates for each prediction, enabling clinicians to identify cases requiring human review. Multi-modal fusion incorporating T2 and FLAIR sequences could provide complementary information to resolve ambiguous cases. Prospective validation in clinical settings with real-time deployment and user studies with radiologists would assess the practical utility and usability of the system.

In conclusion, NeuroDual-Net establishes a new state-of-the-art for simultaneous brain tumor classification and segmentation on the BRISC dataset, achieving 98.60% accuracy while providing Dice scores of 0.905. The multi-task learning framework with Multi-Level Supervision demonstrates that joint optimization of related tasks produces synergistic benefits, improving both tasks relative to single-task baselines.

6. Conclusion

This paper introduced NeuroDual-Net, a multi-task learning framework for simultaneous brain tumor classification and segmentation with inherent interpretability. The architecture employs a shared EfficientNet-B0 encoder with dual decoder heads and Multi-Level Supervision to enhance gradient flow. Comprehensive experiments on the BRISC dataset demonstrate that NeuroDual-Net achieves 98.60% classification accuracy and 0.905 Dice score, outperforming 10 state-of-the-art architectures. Ablation studies confirm that multi-task learning improves accuracy by 6.7% over single-task classification, while MLS contributes an additional 1.2% improvement. Statistical validation confirms the significance of observed improvements. The proposed framework represents a significant advance toward clinically deployable AI systems that provide both diagnostic accuracy and visual explainability.

Author Contributions

Author 1: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Visualization.

Author 2: Conceptualization, Methodology, Writing - review editing, Supervision.

Author 3: Resources, Writing - review editing, Supervision, Project administration, Funding acquisition.

Author 4: Data curation, Resources, Validation.

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